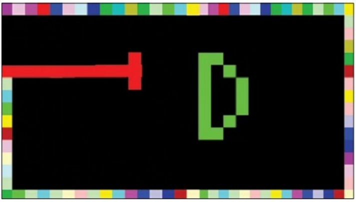

Figure 1: Piet program to print *OSFY*

```
>+++[>++++++++++<-]>++.
>+++++++++[>++++++++++<-]>-.
>+++++++[>++++++++++<-]>+++++++++.
>++++++++[>++++++++++<-]>+++++.
```

The size of the program clearly illustrates the difficulty of writing even the simplest of programs in BF.

Ook!

Ook! is not an independent programming language. It is a derivative language of BF. What makes Ook! interesting is the fact that there are only three keywords -- Ook., Ook?, and Ook!. These three keywords are taken in pairs to form the eight keywords of BF. The three keywords of the language Ook! are created in such a way that they are supposed to be writable and readable even by orangutans. There are some online interpreters freely available for Ook!. The size of programs tends to become even more longer with Ook!. The following is an Ook! program to print my name '*DEEPU*' on the screen.

```
Ook. Ook? Ook. Ook. Ook. Ook. Ook. Ook. Ook. Ook. Ook. Ook.
Ook. Ook. Ook. Ook. Ook. Ook. Ook! Ook? Ook? Ook. Ook. Ook.
Ook. Ook. Ook. Ook. Ook. Ook. Ook. Ook. Ook. Ook. Ook. Ook.
Ook. Ook. Ook. Ook? Ook! Ook! Ook? Ook! Ook? Ook. Ook. Ook.
Ook. Ook. Ook. Ook. Ook. Ook. Ook! Ook. Ook. Ook. Ook! Ook.
Ook! Ook. Ook. Ook? Ook. Ook. Ook. Ook. Ook. Ook. Ook! Ook?
Ook? Ook. Ook. Ook. Ook. Ook. Ook. Ook. Ook. Ook? Ook! Ook!
Ook? Ook! Ook? Ook. Ook. Ook. Ook. Ook. Ook! Ook. Ook. Ook.
Ook. Ook. Ook. Ook. Ook. Ook. Ook. Ook. Ook! Ook.
```

The program is too large for debugging and comprehension; hence, all we can do is hope and pray that the program gives the correct output.

Piet

Piet is an even more bizarre esolang. The programs in Piet do not use any text-based keywords; instead, Piet uses coloured blocks as keywords. The final program obtained in the Piet language will look like an abstract painting. This is the reason why it's named after the Dutch painter Piet Mondarin, who is the pioneer of geometric abstract art. The extension of a Piet program is *.ppm* (Portable Pixelmap Graphics), which is an open graphics file format. The most widely used IDE for Piet is the open source Piet Creator. I tried to create (the word 'write' is not appropriate for Piet programs) a Piet program to print 'OPEN SOURCE FOR YOU'. But after a few hours, I understood that every man has his breaking point; so I satisfied myself with printing '*OSFY*'. Figure 1 is actually the program to print the message '*OSFY*'.

But I also came to know that there are interpreters for Piet created using Piet itself. I was dumbfounded at the mere thought about such a horrendous venture. Imagine the countless agonising hours spent by those geeks to develop these types of near impossible programs. I am sure if there ever were a religion for open source developers, these guys would be canonized as saints by the devotees.

Whitespace

In almost every programming language, whitespace characters are ignored by the compilers and interpreters or have minor implications. But Whitespace is an exception to this rule. It is an esoteric programming language in which the whitespace characters alone are used to generate the program. Every non-whitespace character used in the program is considered as a comment and ignored by the compiler. This language can never be included in the academic curriculum because the answer script containing the correct Whitespace programs will look exactly like the one without any answer – it will be blank! The programs are generated by using spaces, tabs and line feeds alone. The five commands in the language Whitespace are described below. [Space] is used for stack manipulation. The combination [Tab][Space] is used for arithmetic operation. The combination [Tab][Tab] is used for heap access. [Linefeed] is used for flow control, and the combination [Tab][Linefeed] is used for I/O manipulation. Since Whitespace ignores all the non-whitespace characters, programs can be easily embedded inside other free form languages like C. Thus Whitespace can be used to generate *polyglot programs,* which are programs written in a valid form of multiple programming languages.

Java2K

I will end the discussion on esoteric languages with Java2K. It is in no way related to the popular programming language, Java. Most of the languages we ever come across are deterministic programming languages. But Java2K is a probabilistic programming language. Every function in Java2K has two implementations, and the actual version of the function to be executed is selected at runtime, based on the values provided by a pseudo random number generator. So it is extremely difficult to write programs with relatively high accuracy. To make it even more difficult, the language uses a base-11 number system with whitespace as the